Assignment 2

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Data Description:

The purity of oxygen produced by a fractional distillation process is thought to be related to the percentage of hydrocarbons in the main condensor of the processing unit.

a. Enter the data into R and make a dataframe

```
purity <- c(86.91, 89.85, 90.28, 86.34, 92.58, 87.33, 86.29, 91.86, 95.61,
89.86, 96.73, 99.42, 98.66, 96.07, 93.65, 87.31, 95.00, 96.85, 85.20, 90.56)

hydrocarbon <- c(1.02, 1.11, 1.43, 1.11, 1.01, 0.95, 1.11, 0.87, 1.43, 1.02,
1.46, 1.55, 1.55, 1.55, 1.40, 1.15, 1.01, 0.99, 0.95, 0.98)

data <- data.frame(purity, hydrocarbon)

data
```

```
   purity hydrocarbon
1   86.91        1.02
2   89.85        1.11
3   90.28        1.43
4   86.34        1.11
5   92.58        1.01
6   87.33        0.95
7   86.29        1.11
8   91.86        0.87
9   95.61        1.43
10  89.86        1.02
```

```
11  96.73        1.46
12  99.42        1.55
13  98.66        1.55
14  96.07        1.55
15  93.65        1.40
16  87.31        1.15
17  95.00        1.01
18  96.85        0.99
19  85.20        0.95
20  90.56        0.98
```

b. Fit a simple linear regression model to the data

```r
# using in built in function
model <- lm(data$purity ~ data$hydrocarbon)

# computation
n <- length(data$purity)
x <- data$hydrocarbon
y <- data$purity
syy <- sum(y**2) - ((sum(y)**2)/n)
sxx <- sum(x**2) - ((sum(x)**2)/n)
sxy <- sum(x*y) - ((sum(x)*sum(y))/n)

(beta1_hat <- sxy/sxx)
```

```
[1] 11.80103
```

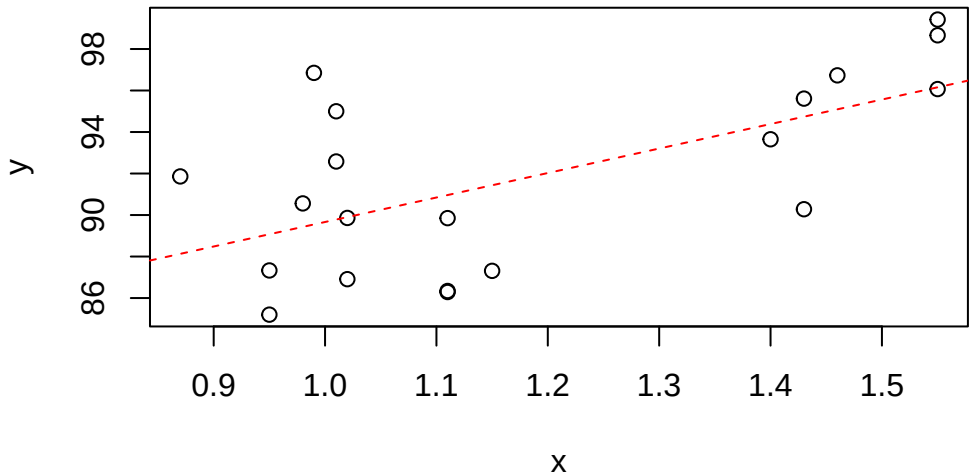
```r
(beta0_hat <- mean(y) - beta1_hat*mean(x))
```

```
[1] 77.86328
```

```r
# to check the coefficients from the model
(model$coefficients)
```

```
     (Intercept) data$hydrocarbon
        77.86328         11.80103
```

```r
plot(x, y)
abline(beta0_hat, beta1_hat, col = "red", lty = 2)
```



c. Calculate R^2 and interpret its value.

```r
# sum square error
(sse <- syy - ((sxy)**2/sxx))
```

```
[1] 232.8344
```

```r
# sum square total
(sst <- syy)
```

```
[1] 381.1473
```

```r
# sum square regression
(ssr <- sst - sse)
```

```
[1] 148.313
```

```r
# rsq (MSE)
(rsq <- ssr/sst)
```

```
[1] 0.3891224
```

> **Note**
>
> R-squared value of 0.39 means the model can explain 39% of the variance. In other words, the variable hydrocarbon can explain 39% of the variability in purity. The remaining 61% of the variance is due to other factors not included in the model or random error.

d. Test the null hypothesis $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$ at $\alpha = 0.01$.

```r
(mse <- sse/(n-2))
```

```
[1] 12.93524
```

```r
(t_0 <- (beta1_hat/sqrt(mse/sxx)))
```

```
[1] 3.386119
```

```r
(p_value <- 2 * (1 - pt(abs(t_0), n-2)))
```

```
[1] 0.003291122
```

```r
cat(p_value>0.01)
```

```
FALSE
```

> **Note**
>
> - The p-value is less than the alpha = 0.01, hence we reject the null hypothesis.
> - This suggests that there is a statistically significant relationship between hydrocarbon and purity.

- beta1 is not equal to zero.

e. Find 95% CI on the slope

```
upper <- beta1_hat + sqrt(mse/sxx) * qt(0.975, n-2)
lower <- beta1_hat - sqrt(mse/sxx) * qt(0.975, n-2)
cat("95% CI for beta1: [", lower, ", ", upper, "]")
```

```
95% CI for beta1: [ 4.479066 ,  19.12299 ]
```

f. Find s 95% CI on the mean purity when the percentage of hydrocarbons is 1.00.

is t with $n-2$ degrees of freedom. Consequently, **a $100(1-\alpha)$ percent CI on the mean response at the point $x = x_0$** is

$$\hat{\mu}_{y|x_0} - t_{\alpha/2,n-2}\sqrt{MS_{\text{Res}}\left(\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)}$$
$$\leq E(y|x_0) \leq \hat{\mu}_{y|x_0} + t_{\alpha/2,n-2}\sqrt{MS_{\text{Res}}\left(\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)} \qquad (2.43)$$

Figure 1: formula: 2.43

```
# using the formula 2.43 from the notes
mu_hat <- beta0_hat + beta1_hat * 1.00
se_mu <- sqrt(mse * (1/n + (1.00 - mean(x))**2/sxx))
upper_mu <- mu_hat + se_mu * qt(0.975, n-2)
lower_mu <- mu_hat - se_mu * qt(0.975, n-2)
cat("95% CI for mean purity when x_0 is 1.00: [", lower_mu, ", ", upper_mu, "]")
```

```
95% CI for mean purity when x_0 is 1.00: [ 87.51017 ,  91.81845 ]
```

g.construct the analysis of variance table

```
# using in built function
summary(aov(model))
```

```
                 Df Sum Sq Mean Sq F value  Pr(>F)
data$hydrocarbon  1  148.3  148.31   11.47 0.00329 **
Residuals        18  232.8   12.94
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# we know ssr, sse, and sst

# degrees of freedom are defined as follows:
df_r <- 1
df_t <- n - 1
df_e <- df_t - df_r

# mean squares
msr <- ssr/df_r
mse <- sse/df_e

# F statistic
(F_stat <- msr/mse)
```

```
[1] 11.4658
```

```r
# p-value for F statistic
(p_value_f <- 1 - pf(F_stat, df_r, df_e))
```

```
[1] 0.003291122
```

```r
# checking hypothesis that at least one predictor is significant
cat(p_value_f < 0.01)
```

```
TRUE
```

> **Note**
>
> - Since the p-value for the F statistic is less than 0.01, we reject the null hypothesis that all regression coefficients are equal to zero.
> - This suggests that hydrocarbon is significantly related to the response variable (purity).